CYBERSRS

Secure Monitoring Platform Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments. The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes. Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary. The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - collect system events, authentication activity, process information, and security alerts from multiple departments' servers and managed endpoints.

Statement: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments' servers and managed endpoints.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A university server and a managed endpoint in operation

When: The monitoring platform initiates data collection

Then: System events, authentication activity, process information, and security alerts are gathered from the devices

Method: test

Observable evidence: Test capture of collected event streams from server and endpoint

Unresolved decisions: Definition of 'unusual login behavior' in context of university user roles

FR-002 - The platform shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Statement: The platform shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A system with normal and anomalous user activities

When: The monitoring platform processes real-time event streams

Then: Unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes are identified and flagged

Method: analysis

Observable evidence: Test instrumented state capture showing detection of anomalous events

Unresolved decisions: Thresholds for defining 'repeated' failed attempts or 'suspicious' process activity

FR-003 - Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Statement: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: A security administrator with authorized access and a test alert from a compromised endpoint

When: The administrator requests alert review and initiates device investigation

Then: The requested alert is returned, investigation notes are recorded, alert status is updated, and the endpoint can be isolated

Method: demonstration

Observable evidence: Test capture of the alert response and endpoint isolation action

Unresolved decisions: Authorization model for changing alert status or isolating endpoints

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Statement: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: An operator and a test alert

When: The operator requests alert review

Then: The requested alert is returned

Method: test

Observable evidence: Test capture of the alert response

Given: A system with multiple campus locations

When: Security events are generated and logged

Then: The system shall maintain audit logs of security events and administrator actions

Method: inspection

Observable evidence: Log entries are stored in a centralized database with timestamp, event type, source, and action

Unresolved decisions: What specific types of events should be logged?; How frequently should logs be retained?; What is the threshold for triggering an alert on unusual login behavior?

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-003

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-003; NFR-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

7. Compliance and policy applicability matrix

Open decisions: Which specific types of events should be logged in the audit logs?; How frequently should logs be retained?; What is the threshold for triggering an alert on unusual login behavior?; Definition of 'unusual login behavior' in context of university user roles; Thresholds for defining ' repeated' failed attempts or 'suspicious' process activity; Authorization model for changing alert status or isolating endpoints

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Security administrators

Detail: Proposed logical endpoint for the source-stated actor: Security administrators.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Component name: Actor: Security administrators

Responsibility: Represent the source-stated actor: Security administrators.

Event Collection Engine

Detail: Proposed logical owner for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-001; NFR-001

Component name: Event Collection Engine

Responsibility: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments' servers and managed endpoints. The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Anomaly Detection Module

Detail: Proposed logical owner for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Component name: Anomaly Detection Module

Responsibility: The platform shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Security Investigation Workflow

Detail: Proposed logical owner for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Component name: Security Investigation Workflow

Responsibility: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

From component: Actor: Security administrators

To component: Security Investigation Workflow

Data or action: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-003.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-003

Untrusted component: Actor: Security administrators

Trusted component: Security Investigation Workflow

Enforcement: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A university server and a managed endpoint in operation
2. When: The monitoring platform initiates data collection

Expected result: Expected: System events, authentication activity, process information, and security alerts are gathered from the devices

Evidence to record: Evidence: Test capture of collected event streams from server and endpoint

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: A system with normal and anomalous user activities
2. When: The monitoring platform processes real-time event streams

Expected result: Expected: Unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes are identified and flagged

Evidence to record: Evidence: Test instrumented state capture showing detection of anomalous events

Test type: analysis

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: A security administrator with authorized access and a test alert from a compromised endpoint
2. When: The administrator requests alert review and initiates device investigation

Expected result: Expected: The requested alert is returned, investigation notes are recorded, alert status is updated, and the endpoint can be isolated

Evidence to record: Evidence: Test capture of the alert response and endpoint isolation action

Test type: positive

Id: TEST-004

Requirement ids: NFR-001

Procedure

1. Given: An operator and a test alert
2. When: The operator requests alert review

Expected result: Expected: The requested alert is returned

Evidence to record: Evidence: Test capture of the alert response

Test type: positive

Id: TEST-005

Requirement ids: NFR-001

Procedure

1. Given: A system with multiple campus locations
2. When: Security events are generated and logged

Expected result: Expected: The system shall maintain audit logs of security events and administrator actions

Evidence to record: Evidence: Log entries are stored in a centralized database with timestamp, event type, source, and action

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-004

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-004

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The decision to select Agile as the required model is based on a uniform score of 40 across all candidates, with no differentiating input evidence provided to favor one methodology over another. Given the absence of explicit factors such as budget, delivery timelines, failure consequences, feedback mechanisms, legacy systems, regulation, schedule constraints, stability needs, uncertainty levels, or verification requirements, no model can be objectively ranked for suitability. The provisional nature of this decision—due to missing evidence and unknown factors—means that the selection remains a baseline assumption without empirical support. All models received equal scores because none of the input evidence differentiated them in terms of alignment with functional (FR) or non-functional (NFR) requirements. The existing requirements emphasize real-time monitoring, alert investigation, and centralized dashboarding, which are well-supported by iterative development cycles in Agile. However, without explicit evidence on feedback loops, regulatory compliance needs, or legacy integration constraints, this choice is not validated against actual operational conditions. Therefore, the selection of Agile remains a default due to lack of distinguishing factors, with all other models equally unproven under current evidence.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments' servers and managed endpoints.; FR-002: The platform shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.; FR-003: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.; NFR-001: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

2. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall collect system events, authentication activity, process information, and security alerts from multiple departments' servers and managed endpoints.; FR-002: The platform shall detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.; FR-003: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.; NFR-001: The system shall maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.
3. In an authorized isolated test environment, execute TEST-001: Given: A university server and a managed endpoint in operation When: The monitoring platform initiates data collection Expected: System events, authentication activity, process information, and security alerts are gathered from the devices
4. In an authorized isolated test environment, execute TEST-002: Given: A system with normal and anomalous user activities When: The monitoring platform processes real-time event streams Expected: Unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes are identified and flagged
5. In an authorized isolated test environment, execute TEST-003: Given: A security administrator with authorized access and a test alert from a compromised endpoint When: The administrator requests alert review and initiates device investigation Expected: The requested alert is returned, investigation notes are recorded, alert status is updated, and the endpoint can be isolated
6. In an authorized isolated test environment, execute TEST-004: Given: An operator and a test alert When: The operator requests alert review Expected: The requested alert is returned

7. In an authorized isolated test environment, execute TEST-005: Given: A system with multiple campus locations When: Security events are generated and logged Expected: The system shall maintain audit logs of security events and administrator actions
8. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
9. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review design and test evidence for violations of source-defined permissions and exclusions.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; NFR-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure monitoring platform for university servers and managed endpoints that collects system events, authentication activity, process information, and security alerts from multiple departments.

Id: E-002

Source: description

Quote: The platform should detect unusual login behavior, repeated failed authentication, suspicious process activity, and unauthorized configuration changes.

Id: E-003

Source: description

Quote: Security administrators should be able to review alerts, investigate affected devices, record investigation notes, change alert status, and isolate compromised endpoints when necessary.

Id: E-004

Source: description

Quote: The system should maintain audit logs of security events and administrator actions and provide a centralized dashboard for monitoring security status across multiple campus locations.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7468

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7429

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-ecd06e0e85ebe229:16 - 03_OWASP_CheatSheets/038_OWASP_Authorization_Cheat_Sheet.md

Source document id: kb-ecd06e0e85ebe229

Chunk index: 16

Page or section: text chunk 17

Relevance score: 0.7322

Sha256: e78a10eb9759a1fc7593ba1fc89a05b017e7e795ac16ad5496bdba86138d891f

kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7183

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.7274

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:19 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 19

Page or section: text chunk 20

Relevance score: 0.7272

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-003	TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-004, TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28